

RADAR Challenge 2026: Robust Audio Deepfake Recognition under Media Transformations

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Abstract—RADAR Challenge 2026 is an APSIPA Grand Challenge on Robust Audio Deepfake Recognition under Media Transformations, designed to simulate realistic media conditions in real-world audio distribution pipelines, including compression, resampling, noise, and reverberation. It consists of two phases: an English development phase with labeled data for analysis and paper writing, and a multilingual evaluation phase containing more than 100,000 utterances in English, Singapore English, Mandarin Chinese, Taiwanese Mandarin, Japanese, and Vietnamese. Systems are evaluated using equal error rate (EER) for binary real/fake classification. This paper describes the challenge task, the construction of the data set, the evaluation protocol, and the overall results. During the challenge, 33 teams submitted to the development phase and 22 teams submitted to the final evaluation phase. The reported results highlight the remaining challenges of robust audio deepfake detection under multilingual and media-transformed conditions.

I. INTRODUCTION

Audio deepfake detection has become an increasingly important research topic due to rapid advances in neural speech synthesis and voice conversion systems. Modern synthetic speech can achieve highly natural quality and can be used in beneficial applications such as accessibility and conversational agents, but it also introduces risks related to impersonation, misinformation, fraud, and attacks on speaker verification systems. To address these concerns, several community benchmarks and challenges, including the ASVspoof [1], [2], [3] series and recent audio deepfake detection evaluations [4], [5], [6], have driven major progress in spoofing countermeasures and evaluation methodologies. However, despite strong performance on many benchmark data sets, robustness under realistic media conditions remains a major open challenge.

In practical deployment scenarios, speech is often compressed [3], resampled, mixed with environmental noise or background music, reverberated [7], edited [8], or processed through real-world media pipelines before analysis. These transformations can suppress or distort spoofing artifacts while also introducing non-spoof artifacts that affect detector behavior. Existing systems may therefore achieve strong performance under clean conditions while remaining vulnerable to distribution shifts caused by media transformations, multilingual conditions, unseen spoofing sources, and limited development time constraints during challenge participation.

To study these issues, the APSIPA RADAR Challenge 2026 focuses on Robust Audio Deepfake Recognition under Media Transformations¹. Unlike benchmarks centered mainly on clean synthetic speech, we emphasize robustness under realistic media processing conditions. The challenge consists of two phases: an English development phase with labeled data for system analysis and paper writing, and a multilingual evaluation phase containing more than utterances across English, Singapore English, Mandarin Chinese, Taiwanese Mandarin, Japanese, and Vietnamese. Systems are evaluated using equal error rate (EER) for binary bonafide/spoof classification.

This paper presents the challenge task, the data set construction process, media transformation design, the evaluation protocol, the participant systems, and the overall results. During the challenge, 33 teams submitted results to the development phase and 22 teams submitted results to the evaluation phase. The reported results highlight the remaining challenges of robust audio deepfake detection under multilingual and media-transformed conditions, including transformation mismatch, spoof-source variability, cross-language generalization, and robustness under practical development constraints.

II. CHALLENGE OVERVIEW

Task Definition: The RADAR 2026 challenge focuses on binary real/fake classification for audio deepfake detection. Each participating team is provided with a speech dataset and required to submit a score file indicating whether each utterance is bona fide or spoofed.

Metrics: System performance is evaluated using Equal Error Rate (EER), where lower values indicate better detection performance. The final leaderboard ranking is determined solely based on the Phase 2 evaluation results.

Challenge Motivation: Unlike previous audio deepfake detection challenges that primarily evaluate systems on relatively clean or lightly processed speech, the RADAR 2026 challenge emphasizes robustness under realistic media-transformed conditions. In practical deployment scenarios, speech recordings are often affected by codec compression, resampling, bandwidth limitation, additive noise, background music, reverberation, loudness normalization, dynamic range compression, trimming, and other distortions introduced during recording,

¹ <https://radar-challenge.github.io/>

TABLE I
LANGUAGE DISTRIBUTION OF THE EVALUATION DATASET.

Language	Code	Bonafide	Spoof	Total	Spoof %	Dataset %
English	en	15,000	17,923	32,923	54.44%	32.05%
Singapore English	en-sg	3,500	6,711	10,211	65.72%	9.94%
Mandarin	zh	11,500	11,234	22,734	49.41%	22.13%
Taiwanese Mandarin	zh-tw	3,500	1,678	5,178	32.41%	5.04%
Japanese	ja	9,500	8,742	18,242	47.92%	17.76%
Vietnamese	vi	7,000	6,438	13,438	47.91%	13.08%
Total	—	50,000	52,726	102,726	51.33%	100.00%

transmission, editing, or social media sharing. To address this gap, the challenge evaluates both bona fide and spoofed speech under diverse and unseen media transformations, with a particular focus on robustness and generalization.

Training Policy: we adopt an open-training policy, where participants are free to construct their own training pipelines using publicly accessible data, provided that data overlapping with the challenge development set is excluded. This better reflects practical deployment scenarios and encourages methodological diversity across training data, multilingual coverage, augmentation strategies, pretrained models, calibration methods, and score fusion approaches, while maintaining fixed development and evaluation sets for fair comparison. The open-training design also allows participating teams to leverage their individual strengths, resources, and research expertise, resulting in a diverse range of approaches contributed by teams from different institutions and regions worldwide.

Phase 1 (Development): supports controlled system analysis, error inspection, hyperparameter tuning, and paper preparation. The development set consists of English speech derived from existing audio deepfake datasets and includes media-transformed bona fide and spoofed utterances. The development phase was conducted over 11 days, during which each team was allowed two submissions per day. Ground-truth labels were released right after to facilitate post-hoc analysis and discussion in technical reports and challenge papers.

Phase 2 (Evaluation): assesses robustness and generalization under challenging multilingual and media-transformed conditions. The evaluation set contains more than 100,000 utterances in English, Singapore English, Mandarin Chinese, Taiwanese Mandarin, Japanese, and Vietnamese. Compared to the development phase, the evaluation phase introduces greater variability in language, spoofing conditions, and media transformation combinations. During the challenge, participants were informed only that the evaluation data included four languages, without disclosure of the regional dialect variations. The evaluation phase was conducted over 8 days, during which each team was allowed one submission per day.

III. DATASET CONSTRUCTION

A. Development Set

The development set is derived from the LlamaPartial-Spoof [8] full-fake subset with additional media transformations applied to both bona fide and spoofed speech. In total, the set contains 44,034 English utterances, including 10,573

bona fide sourced from the LibriTTS [9] development and test partitions and 33,461 synthetic utterances generated using six speech synthesis systems: LJ JETS [10], YourTTS [11], XTTS-v2 [12], GPT-SoVITS [13], ElevenLabs [14], and CosyVoice 1.0 [15]. By extending an existing public audio deepfake dataset with additional media degradations, the development condition provides an interpretable benchmark for controlled robustness analysis while remaining distinct from the final blind evaluation condition. Since both the original LlamaPartialSpoon [8] dataset and the RADAR2026 development set and their labels are available after the completion of Phase 1, participants can perform detailed diagnostic analyses, including error inspection, score-distribution analysis, ablation studies, robustness analysis, and hypothesis testing. Therefore, the development set by itself is a meaningful robustness benchmark.

B. Evaluation Set

The evaluation set was specifically developed as a new benchmark for robust audio deepfake detection under realistic multilingual and media-transformed conditions. Rather than maximizing language coverage, the benchmark focuses on several major language conditions relevant to the Asia-Pacific region: English (en), Singapore English (en-sg), Mandarin Chinese (zh), Taiwanese Mandarin (zh-tw), Japanese (ja), and Vietnamese (vi). Table I summarizes the language and class distribution. The final evaluation set contains 102,726 utterances, including 50,000 bona fide and 52,726 spoofed samples, resulting in a near-balanced overall class distribution. English is the largest language condition (32.05%), followed by Mandarin (22.13%), Japanese (17.76%), and Vietnamese (13.08%), while Singapore English and Taiwanese Mandarin account for 9.94% and 5.04% of the dataset, respectively. Although the overall dataset is near-balanced, class balance varies across language conditions, with Singapore English being spoof-heavy and Taiwanese Mandarin being bona fide-heavy. Since EER is threshold-based, language-dependent score distributions may affect global calibration.

The benchmark was deliberately designed to include multiple languages, bona fide sources, spoofing systems, and media-transformation conditions without enforcing uniform distributions across all factors. This better reflects practical deployment conditions, where detectors encounter uneven language distributions, heterogeneous recording sources, diverse synthesis systems, and unpredictable processing pipelines.

TABLE II
BONAFIDE SPEECH SOURCES USED IN THE EVALUATION DATASET.

Bonafide Source	en	en-sg	zh	zh-tw	ja	vi	Total	Subset %
Common Voice Scripted Speech [16]	8,000	1,500	5,000	2,000	6,000	3,000	25,500	51.00%
People’s Speech [17]	7,000	—	—	—	—	—	7,000	14.00%
IMDA [18]	—	2,000	—	—	—	—	2,000	4.00%
MAGICDATA Mandarin Read Speech [19]	—	—	6,500	—	—	—	6,500	13.00%
FormosaSpeech [20]	—	—	—	1,500	—	—	1,500	3.00%
CPJD [21]	—	—	—	—	3,500	—	3,500	7.00%
FOSD [22]	—	—	—	—	—	4,000	4,000	8.00%
Total	15,000	3,500	11,500	3,500	9,500	7,000	50,000	100.00%

TABLE III
SPOOF SPEECH SOURCES USED IN THE EVALUATION DATASET.

Text-to-Speech System	Type	en	en-sg	zh	zh-tw	ja	vi	Total	Subset %
iFlytek ¹	Commercial	1,885	—	851	—	313	1,474	4,523	8.58%
Houshan ²	Commercial	1,678	—	989	—	1,502	—	4,169	7.91%
ElevenLabs ³	Commercial	1,794	539	1,221	340	833	618	5,345	10.14%
Cartesia ⁴	Commercial	1,609	—	809	—	933	868	4,219	8.00%
OpenAI ⁵	Commercial	876	—	1,415	—	717	1,187	4,195	7.96%
Chatterbox ⁶	Open-source	1,884	2,915	1,371	—	939	—	7,109	13.48%
CosyVoice 3.0 [23]	Open-source	1,770	2,688	1,098	348	1,371	—	7,275	13.80%
Qwen3-TTS [24]	Open-source	2,204	—	1,280	336	1,060	—	4,880	9.26%
Fish Audio S2 Pro [25]	Open-source	1,232	569	1,592	654	1,074	841	5,962	11.31%
Piper ⁷	Open-source	2,991	—	608	—	—	1,450	5,049	9.58%
Total	—	17,923	6,711	11,234	1,678	8,742	6,438	52,726	100.00%

¹ https://www.xfyun.cn/services/online_tts ² <https://www.volcengine.com/product/tts> ³ <https://elevenlabs.io/> ⁴ <https://cartesia.ai>
⁵ <https://openai.com/> ⁶ <https://github.com/resemble-ai/chatterbox> ⁷ <https://github.com/OHF-Voice/piper1-gpl>

1) *Bona fide subset*: The bona fide subset contains 50,000 utterances collected from multiple multilingual speech corpora. Table II summarizes the bona fide data sources. The largest source is Common Voice Scripted Speech, which contributes 25,500 utterances (51.00% of the bona fide subset) spanning all target language conditions. Since Common Voice is a widely used public-domain speech dataset, we assumed that many participants may have already incorporated portions of it into their training pipelines. Additional sources were therefore included to provide broader language-specific coverage across English, Singapore English, Mandarin Chinese, Taiwanese Mandarin, Japanese, and Vietnamese speech conditions.

2) *Spoof Subset*: The spoof subset contains 52,726 utterances generated using ten speech synthesis systems. Table III provides the source-level breakdown. It was intentionally designed to be diverse, covering commercial, open-source, and research-oriented synthesis systems. For commercial services, speech was generated using voices provided directly by the respective platforms. In contrast, open-source systems employed varying voice-cloning methodologies depending on the capabilities and design of each tool. This diversity reduces the likelihood that detectors can rely on artifacts from a single generator and instead encourages the development of features that generalize across synthesis families, architectures, and languages.

IV. MEDIA TRANSFORMATION

RADAR Challenge 2026 focuses on robust audio deepfake detection under realistic media-transformed conditions. Both the development and evaluation sets apply diverse media

transformation pipelines to bona fide and spoofed speech to simulate practical recording, editing, transmission, and distribution workflows. Tables IV and V summarize the transformation pipelines used for the development and evaluation sets, respectively. During data generation, each utterance passes through the entire transformation pipeline (in top to bottom order), where individual transformations are applied probabilistically according to their configured activation chances. As a result, a sample may undergo none, one, or multiple sequential transformations within a single processing chain. In several stages, multiple related transformations are grouped so that only one transformation from the group is applied. For example, codec compression stages randomly select a single codec condition (e.g., Opus, MP3, or AAC) when activated.

The development pipeline applies a moderate set of transformations intended to support controlled robustness analysis and system development. These include silence trimming, zero padding, room impulse response (RIR) reverberation, additive noise, background music, dynamic range compression, fade-in/fade-out effects, peak-level normalization, and codec compression using Opus and MP3. The transformation probabilities and parameter ranges were intentionally constrained to maintain relatively interpretable conditions while still introducing meaningful distribution shifts compared with clean speech.

In contrast, the evaluation pipeline introduces substantially greater diversity and variability in both transformation types and parameter ranges. In addition to the transformations used in the development set, the evaluation condition includes environmental sound events, bandwidth limitation, loudness normalization, resampling to 8.kHz, AAC compression, streaming

TABLE IV
MEDIA TRANSFORMATION PIPELINE USED FOR THE DEVELOPMENT SET.

Chance	Transformation
50%	Silence trimming (leading: 30–80%, trailing: 30–90%)
25%	RIR reverberation (MIT RIR dataset [26])
50%	Zero padding (leading: 0–250 ms, trailing: 0–400 ms)
25%	Background noise (SNR: 15–25 dB, MUSAN noise [27])
	Background music (SNR: 12–22 dB, FMA small [28])
15%	Dynamic range compression
25%	Fade-in/fade-out (fade-in: 20–80 ms; fade-out: 50–150 ms)
60%	Peak-level normalization (target peak: −12 to 1 dBFS)
50%	Opus codec (32–128 kbps)
	MP3 codec (64–160 kbps)

TABLE V
MEDIA TRANSFORMATION PIPELINE USED FOR THE EVALUATION SET.

Chance	Transformation
75%	Silence trimming (leading & trailing: 50–100%)
20%	RIR reverberation (Aachen RIRs [29], Simulated RIRs [30], Synthetic RIRs [7])
20%	Zero padding (leading: 0–400 ms, trailing: 0–500 ms)
20%	Background music (SNR: 8–20 dB, FMA small [28])
20%	Sound Effect (SNR: 4–12 dB, BSD10k sound effect [31])
10%	Bandwidth limitation (passband: 300–3400 Hz;)
20%	Dynamic range compression
20%	Fade-in/fade-out (fade-in: 20–80 ms; fade-out: 50–150 ms)
100%	Peak-level normalization (target peak: −12 to 1 dBFS)
20%	Loudness normalization (target: −18 to −14 LUFS)
10%	Resampling to 8 kHz (final output formatted at 16 kHz)
	Opus codec (24–96 kbps)
50%	MP3 codec (48–128 kbps)
	AAC codec (64–96 kbps)
10%	Streaming dropout simulation (frame: 10–40 ms; dropout enter probability: 2%–8%)
10%	Speech perturbation (speed factor: 0.95–1.05)

dropout simulation, and speech perturbation. Several transformations also use more aggressive parameter ranges, such as stronger silence trimming, lower signal-to-noise ratios, and lower codec bitrates. Multiple transformations may be applied sequentially within a single processing chain, thereby simulating realistic media-processing pipelines encountered in practical deployment scenarios.

The transformation design reflects the observation that real-world speech recordings often undergo multiple stages of processing before reaching an audio deepfake detector. Speech may be edited, normalized, compressed by communication platforms, resampled by applications or devices, mixed with environmental sounds or background music, or affected by room acoustics and transmission artifacts. These transformations can either suppress spoofing artifacts or introduce confounding distortions. For example, codec compression may remove high-frequency synthesis artifacts, while reverberation and environmental interference can obscure temporal fine structure. Consequently, the challenge encourages the development of systems that rely on robust and generalizable evidence rather than narrow generator-specific artifacts.

V. BASELINE

We provide a single baseline system based on the open-source SSL-AASIST anti-spoofing model [32], which com-

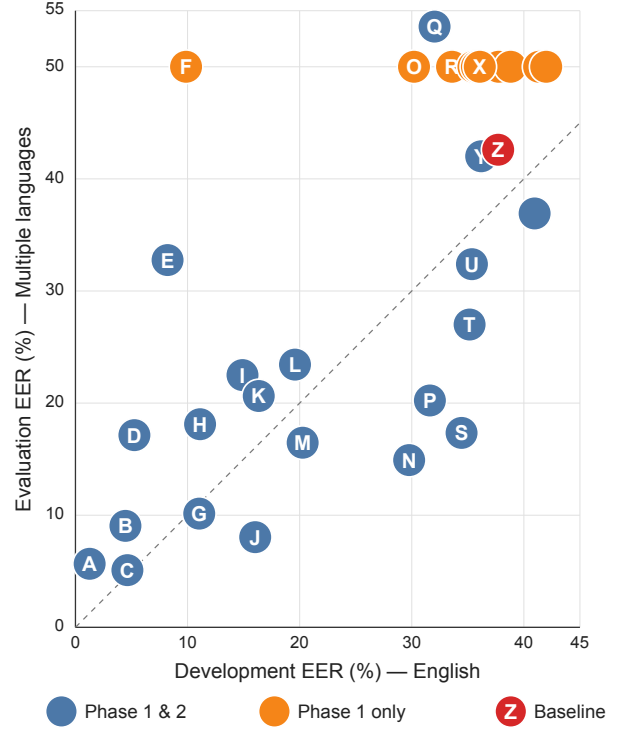


Fig. 1. The EER results in Phase 1 and Phase 2 of the top 26 teams.

bines a wav2vec 2.0 frontend with the AASIST backend. Despite its moderate size of approximately 318M parameters, the model serves as a strong and accessible reference system. We directly use the pretrained model released by the original authors and provide example evaluation scripts and score analysis [33] for reproducibility and easy onboarding. Since the model was originally trained on ASVspoof 2019 LA [1] with RawBoost-based augmentation, it is not optimized for the multilingual and media-transformed conditions of RADAR 2026. The baseline achieved 37.71% EER on the Phase 1 development set and 42.6% EER on the multilingual Phase 2 evaluation set. The baseline is intended as a reference system rather than a performance ceiling, and participants are encouraged to improve robustness through stronger augmentation, multilingual training data, calibration, model fusion, and alternative architectures. During Phase 1, teams were required to outperform the baseline on the development set in at least one valid submission to qualify for the evaluation phase.

VI. RESULTS

Figure 1 shows the EER results of the top 26 teams. Each team is assigned an ID from A to Z according to its Phase 1 ranking, with Z corresponding to the baseline. Among the 26 systems, 20 teams participated in the evaluation phase, while 6 teams did not for various reasons; these systems are assigned an evaluation EER of 50% in the figure. Team A achieved the best result on the Phase 1 development set with 1.27% EER, while Team C achieved the best result on the multilingual Phase 2 evaluation set with 5.10% EER.

The leaderboards show that top-ranked systems achieved substantially lower EERs in both phases, indicating that the task is challenging but solvable with sufficiently robust approaches. Performance differences likely reflect variations in model design, training data, augmentation strategies, and score calibration. Since the development labels were released after Phase 1, the development leaderboard should be interpreted as a controlled released-label benchmark, while the final evaluation leaderboard serves as the primary assessment of blind generalization under realistic multilingual and media-transformed conditions. As RADAR Challenge 2026 adopts an open-training setting, we look forward to teams sharing their approaches, system designs, and data usage in their technical reports. Additional analyses and system-level comparisons will be reported when more information becomes available.

VII. DISCUSSION

A. Limitations of Simulated Media Transformation

Although the challenge introduces diverse media transformations to approximate realistic deployment conditions, the pipeline remains simulation-based and may not fully reflect real-world media processing [34]. In practice, audio shared through communication platforms, social media, recording devices, and editing applications may undergo more complex or proprietary processing chains that are difficult to reproduce in a controlled benchmark [35]. Another limitation is the assumption that transformed bona fide speech should remain labeled as bona fide. In practice, severe media processing may introduce artifacts that overlap with spoof-related characteristics, especially under low-quality or heavily degraded conditions. Nevertheless, for simplicity and consistency during the challenge, all transformed utterances retain their labels.

B. Limitations of the Current Analysis

The current analysis is primarily based on aggregate leaderboard results and dataset-composition statistics, and therefore does not yet provide detailed condition-level or system-level analysis. Since many participant system descriptions and training details are not yet publicly available, it remains difficult to determine which specific language conditions, media transformations, spoofing systems, model architectures, augmentation strategies, or training data contributed most to robustness and generalization performance. We therefore plan to prepare a more comprehensive follow-up analysis after additional participant technical reports and system-description papers become available.

VIII. CONCLUSION

This paper summarized the APSIPA RADAR Challenge 2026, a new benchmark for robust audio deepfake detection under realistic media transformations. The challenge includes an English development set for controlled analysis and a large-scale blind evaluation set spanning six language conditions and diverse media-processing pipelines. The results show that robust audio deepfake detection under multilingual and media-transformed conditions remains challenging, although

several teams achieved promising performance despite the limited development timeline and blind evaluation setting. RADAR Challenge 2026 is intended to support future research on media-robust audio deepfake detection, including transformation-aware training, multilingual modeling, calibration, and robustness evaluation under realistic deployment conditions. Both the development and evaluation sets will be publicly released after APSIPA 2026, and future work will include more detailed condition-level and system-level analysis as additional participant reports become available.

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